

MODULE 5.3.1 — BIOEQUIVALENCE STUDY REPORT SYNOPSIS

AMLOZA (Amlodipine Besylate Tablets, USP) 10 mg vs. Norvasc® (Amlodipine Besylate) 10 mg Tablets

IMPORTANT: All subject counts, PK parameters, and results in this document are ILLUSTRATIVE / HYPOTHETICAL, constructed for academic portfolio demonstration only. No real clinical trial was conducted. Do not cite as real data.

Part A: Comparative In Vitro Dissolution Data

Comparative dissolution testing was conducted to support the formulation development rationale described in Module 3.2.P.2 and to characterize in vitro release behavior relative to the RLD prior to conduct of the in vivo bioequivalence study.

Test Conditions

Apparatus	USP Apparatus II (Paddle), 50 rpm
Media	0.1N HCl (pH 1.2), Acetate Buffer (pH 4.5), Phosphate Buffer (pH 6.8), 900 mL, 37°C ± 0.5°C
Sampling timepoints	5, 10, 15, 20, 30, 45, 60 minutes
Batches compared	AMLOZA 10 mg (exhibit batch) vs. Norvasc® 10 mg (reference, procured from US market)

Illustrative Results (f2 Similarity Factor)

pH 1.2	f2 = 68 (similar; f2 ≥ 50 indicates similarity)
pH 4.5	f2 = 71 (similar)
pH 6.8	f2 = 65 (similar)

Illustrative data indicate both test and reference products release ≥85% of label claim within 30 minutes across all three media, consistent with rapid dissolution behavior expected for a highly soluble, highly permeable compound as reported in the literature for amlodipine, formulated by direct compression.

Part B: In Vivo Bioequivalence Study Report — Synopsis

Title

A Randomized, Open-Label, Single-Dose, Two-Treatment, Two-Period, Two-Sequence Crossover Bioequivalence Study of AMLOZA (Amlodipine Besylate) 10 mg Tablets and Norvasc® (Amlodipine Besylate) 10 mg Tablets in Healthy Adult Human Subjects Under Fasting Conditions.

Objective

To assess the bioequivalence of AMLOZA 10 mg Tablets (Test) with Norvasc® 10 mg Tablets (Reference) following a single oral dose under fasting conditions, per FDA Product-Specific Guidance for Amlodipine Besylate.

Study Design

Design	Randomized, open-label, single-dose, two-treatment, two-period, two-sequence, crossover design
Subjects (planned)	36 healthy adult subjects (illustrative; final N accounts for anticipated dropout, powered per intra-subject CV from published literature)
Washout period	≥21 days between periods (justified by amlodipine's long elimination half-life of ~30–50 hours, ensuring full washout ≥ 5 half-lives)
Dosing condition	Fasting (≥10-hour overnight fast prior to dosing)
Dose administered	One tablet (10 mg) with 240 mL water
Sampling schedule	Pre-dose (0h) and post-dose at 0.5, 1, 1.5, 2, 2.5, 3, 4, 6, 8, 12, 24, 48, 72, 96, 120, and 144 hours (extended schedule to characterize the terminal elimination phase given the long half-life)
Bioanalytical method	Validated LC-MS/MS method for plasma amlodipine quantification, per FDA Bioanalytical Method Validation Guidance

Pharmacokinetic Parameters Assessed

C_{max}, T_{max}, AUC_{0–t}, AUC_{0–∞}, t_{1/2}, K_{el}

Statistical Analysis

C_{max} and AUC_{0–t} were analyzed using ANOVA on natural-log-transformed data. Bioequivalence was to be concluded if the 90% confidence interval for the ratio of geometric least-squares means (Test/Reference) for C_{max} and AUC_{0–t} fell within 80.00–125.00%, per FDA guidance.

Illustrative PK Results Summary

Parameter	Test (AMLOZA) vs. Reference (Norvasc) — Geometric Mean Ratio (90% CI)
C_{max}	104.2% (96.8–112.1%)
AUC_{0–t}	101.6% (95.3–108.3%)
AUC_{0–∞}	101.9% (95.0–109.3%)

Illustrative results (all figures hypothetical) fall entirely within the 80.00–125.00% acceptance range for both C_{max} and AUC_{0–t}. If comparable results were obtained in an actual adequately powered study conducted in accordance with FDA requirements, such results would be expected to support a conclusion of bioequivalence between AMLOZA 10 mg Tablets and Norvasc® 10 mg Tablets under the studied conditions.

Safety

In a real study, adverse events would be monitored throughout via clinical observation, vital signs, and subject-reported symptoms, with any events graded for severity and causality. No safety data is presented here as no real study was conducted.

Conclusion

The illustrative pharmacokinetic results presented are hypothetical and are included solely to demonstrate understanding of FDA bioequivalence study design, statistical analysis, and acceptance criteria; no real bioequivalence study was conducted. If comparable results were obtained in an actual adequately powered study conducted in accordance with FDA requirements, such results would be expected to support a conclusion of bioequivalence between AMLOZA (Amlodipine Besylate) 10 mg Tablets and Norvasc® (Amlodipine Besylate) 10 mg Tablets under fasting conditions. A separate bioequivalence study, or a biowaiver extension for additional strengths per 21 CFR 320.22(d)(2) and applicable FDA guidance (supported by proportional formulation and comparative dissolution), would typically be used to justify the 5 mg strength without a second full in vivo study.

Portfolio Note: This synopsis demonstrates understanding of FDA bioequivalence study design requirements for amlodipine (fasting-only design, extended sampling due to long half-life, 90% CI acceptance criteria) and of the biowaiver-extension logic used to avoid duplicating a full BE study across multiple strengths of the same formulation — a real and commonly tested regulatory concept (SUPAC-IR / biowaiver for additional strengths, 21 CFR 320.22(d)(2)).

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